PRANAV MILIND BIDVE

(646) 765-2125 | pranavbidve12@gmail.com | linkedin.com/in/pranavbidve | https://pranavbidve.github.io/

WORK EXPERIENCE

Qosmic AI New York, NY
Founding AI Engineer *Jun 2026 - Present*

- Delivered 126%+ average revenue growth across 300+ stores by building a Deep Agents-powered e-commerce optimization platform using LangGraph, Playwright, and Shopify data to analyze 500 storefront checkpoints.
- Owned LangSmith LLM evaluations, improving recommendation quality by 20% through evaluators and multi-turn tests for grounding, opportunity sizing, and faster fixes
- Reduced audit-to-experiment time by 30% by automating storefront analysis, variant generation, and approval-ready Shopify previews using Next.js and Supabase

TD Bank New York, NY
AI Intern *Sep 2025 - Dec 2025*

- Built an agentic RAG-based fraud detection system using LangGraph + LLM reasoning, reducing false negatives by 10% and automating investigation workflows across 10K transactions/day
- Cut analyst review time by 30% by integrating OLMo-based LLM reasoning into flagged transaction explanations, generating natural language justifications for model decisions via FastAPI
- Optimized detection F1 by 18% by engineering a feature pipeline with SpaCy, and contextual embeddings from transaction text, advancing from logistic regression to XGBoost

Kordis New York, NY
AI Intern *Jun 2025 - Sept 2025*

- Decreased manual review effort by 30% by deploying a hybrid RAG pipeline for transaction summarization and categorization, combining dense retrieval with BM25 over a Pinecone index
- Designed a financial analyst agent and cut p95 latency by 25% with an intelligent query cache on Redis, enabling real-time LLM calls to live financial data
- Enhanced cash flow forecasting accuracy by 12% by training XGBoost, NeuralProphet, and Transformer models on transaction sequences capturing seasonality and long-term trend patterns

PUBLICATION

SLiM-Eval [Under Review, ACL ARR] *Mar 2026*

- Evaluated post-training quantization effects on small language models across 13 model-precision configurations, 5 architectures, and 3 quantization methods - GPTQ, AWQ, and SmoothQuant on NVIDIA A100 and RTX 4090
- Found reasoning tasks degrade 3 to 10x more than factual tasks under INT4 quantization across all tested architectures, surfacing compression tradeoffs critical for production LLM deployment

PROJECTS

HexaNote *Feb 2026*

- Delivered 60% faster AI response times vs CPU-only by enabling Snapdragon NPU acceleration in a fully local, privacy-first note-taking app built with on-device LLMs and FAISS-backed semantic search
- Built a hybrid RAG pipeline with embedding-based retrieval, cross-encoder reranking, and a local LLM to surface context-aware answers from personal notes without any cloud calls

SKILLS

Languages & Tools: Python, SQL, R, C++, TypeScript, Bash, Git, ETL
AI/ML Frameworks: PyTorch, TensorFlow, Scikit-Learn, Hugging Face, XGBoost, ONNX
Libraries & Platforms: Pandas, NumPy, D3.js, SpaCy, Presidio, Retool
LLM Stack: LangChain, LangGraph, RAG, Fine-tuning, Embeddings, FAISS, Pinecone, ChromaDB, Reranking
Cloud & Technologies: GCP, AWS, Snowflake, Hadoop, Spark, SDLC, Postgres, Database Management, Web Scraping, MLflow, FastAPI, Node.js, Airflow, CI/CD

EDUCATION

Columbia University New York, NY
Master of Science, Data Science
Coursework: DL for NLP, LLM based Generative AI, Scaling LLMs, Machine Learning, Statistical Inference
Teaching Assistant: Neural Networks & Deep Learning, Computer Graphics, Business Analytics

Vellore Institute of Technology Vellore, India
Bachelor of Technology, Computer Science and Engineering